Garmin to OSM Sync - One-Page App Summary

What it is

A Python CLI that syncs Garmin Connect activities to OpenStreetMap as GPX traces.
It uses OSM OAuth2 tokens and a local SQLite record to prevent duplicate uploads.

Who it is for

Primary persona: individual athletes/mappers using Garmin Connect who want OSM trace upload automation.

What it does

- Loads Garmin and OSM settings from .env (credentials, OAuth client, behavior flags).
- Runs browser-based OSM OAuth2 authorization and stores tokens in tokens.json.
- Refreshes expired OSM access tokens automatically via refresh_token.
- Logs into Garmin Connect and fetches recent activities (MAX_ACTIVITIES).
- Downloads each new activity GPX into downloads/ for upload processing.
- Uploads GPX files to OSM /api/0.6/gpx/create with bearer authentication.
- Records processed activity IDs in data.db (processed_activities) to avoid re-uploads.

How it works (repo-evidence architecture)

Components: main.py orchestrator; OAuth callback HTTP server; SQLite state store.
Services: Garmin Connect API (garminconnect lib), OSM OAuth2/token and GPX upload APIs.
Data flow: .env -> authorize/refresh -> fetch Garmin activities -> download GPX
-> upload to OSM -> mark activity as processed in data.db -> delete temp GPX.
Not found in repo: packaged daemon/service deployment configuration.

How to run (minimal)

1. cp env_example .env and fill GARMIN_EMAIL, GARMIN_PASSWORD, OSM_CLIENT_ID, OSM_CLIENT_SECRET, REDIRECT_URI (README).
2. python -m venv .venv && source .venv/bin/activate
3. pip install -r requirements.txt
4. python main.py
5. First run: approve OSM OAuth in browser; tokens are saved to tokens.json.